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324813 (24)

BE (8th Semester)

Examination, April-May, 2013

Branch : Electrical

INSTALLATION MAINTENANCE AND TESTING OF ELECTRICAL EQUIPMENTS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each unit carrying 2 marks is compulsory. Attempt any two parts out of (b), (c) and (d).

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Unit - I

Q. 1. (a) Name four electrical safety appliances.

(b) What are the different activities from receipt of equipment to commissioning at site?

(c) Explain maintenance practices followed in the industries. What is the objective of maintenance?

(d) (i) Why equipment/electrical appliance earthing is necessary?

(ii) Explain earthing system in EHT outdoor yard.

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Unit - II

Q. 2. (a) What are the different methods of cooling of transformers?

(b) What are the causes of low value of insulation resistance of transformer? How can the insulation be brought to required level?

(c) Give details of routine tests and type tests to be conducted on large capacity transformers. What is the significance of each test?

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(d) Explain routine maintenance procedure for transformers up to 1000 KVA capacity.

Unit - III

Q. 3. (a) Which type of circuit breakers are used for low voltage/high voltage/extra high voltage applications?

(b) With a neat diagram, explain various equipment on HT and LT side in a transformer substation along with their function.

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(c) (i) What is fault clearing time, relay time and circuit breaker total break time?

(ii) Explain trip circuit and automatic operation of circuit breaker on fault.

(d) State the possible trouble, causes and remedial actions for circuit breakers (outdoor).

Unit - IV

Q. 4. (a) State the factors to be considered for selecting a motor for a particular application.

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(b) Explain the need of cooling system for electrical motors. Explain the principle of cooling system and different mediums.

(c) What are the common problems in a.c. and d.c. motors? How those are rectified?

(d) Give a list of tests conducted at site during installation of medium/large capacity induction motors.

Unit - V

Q. 5. (a) Define hot line maintenance.

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(b) Why hot line maintenance is necessary? What are its advantages and disadvantages?

(c) Explain hot line washing. Why is it required?

(d) Discuss special tools required for hot line maintenance.

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